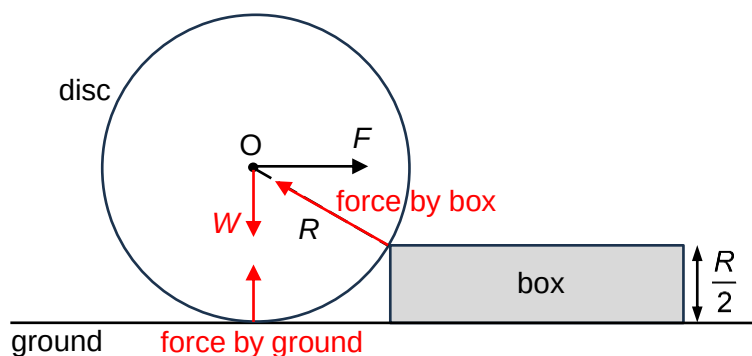


3 (a)



When the disc is just about to rotate, the contact force by the ground just becomes zero.

Perpendicular distance from corner of box to line-of-action of  $F$  is  $\frac{R}{2}$ .

Perpendicular distance from corner of box to line-of-action of  $W$  is  $\sqrt{R^2 - \left(\frac{R}{2}\right)^2} = \frac{\sqrt{3}}{2}R$ .

Applying the principle of moments about the corner of box,

$$F \times \frac{R}{2} = W \times \frac{\sqrt{3}}{2}R$$

$$\frac{F}{W} = \sqrt{3} = 1.7321 = 1.73$$

- (b)  $F$  acting at  $O$  needs to be inclined upwards such that it is at an angle above the horizontal to produce a clockwise moment about the corner to overcome the anticlockwise moment due to the weight.

OR

$F$  acting at  $O$  needs to be inclined upwards so that there is an upward vertical component to produce a clockwise moment about the corner to overcome the anticlockwise moment due to the weight.

OR

$F$  needs to be shifted upwards above  $O$  so that there is a moment arm from the corner of the box to the line-of-action of  $F$ , to produce a clockwise moment about the corner to overcome the anticlockwise moment due to the weight.

Note: Increasing the magnitude of the horizontal force  $F$  acting at  $O$  will not cause any rotation as  $F$  has no moment about the corner of the box because the perpendicular distance from the corner to the line-of-action of  $F$  is zero.

**Raffles Institution**  
**Year 5-6 Physics Department**

4 (a) (i) At the top of the circle